

THOMAS ROBERT MALTHUS

&

DAVID RICARDO



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Thomas Robert Malthus & David Ricardo

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Thomas Robert Malthus (1766-1834)

Thomas Robert Malthus was born in a comfortably well-to-do English family and educated at Cambridge University. As Malthus lived in tumultuous times of intense class conflicts, he pointed out his position on these conflicts in his writings.

He mainly focused on two types of conflicts:

1. Conflicts between the workers and capitalists.
2. Conflicts between the landlord and the capitalists.

Conflicts between the workers and the capitalists

The Industrial Revolution gave a pace in human productivity continuing the process of manufacturing commodities with the widespread construction of factories and the extensive use of machinery and mechanical intelligence. At that time, people with low economic and political power sacrificed much to cope with that pace of industrialization. In 1750, the working class was living at a subsistence level and it deteriorated in the second half of the 18th century.

Poor became poorer, rich and the middle class grew wealthier

Due to the new factory system, there began a lot of struggles and complexity in the workers' lives and the working system. Videlicet,

1. Workers lost their pride in workmanship as their handicraft industries did not last.
2. They lost their direct access to the means of production and had to sell their labor power depending on the market conditions.
3. They lost the audacity to do the job as they did before, pace of work was determined by the machine while maintaining regularity.
4. They became mere appendages to the implacable and pacesetting machine.
5. The extensive division of labor made some rigorous situations that employed women and children with much lower wages than men.

6. Women and children worked to have a daily meal which was insufficient but mandatory for living a life for the whole family. They could easily be managed better than men for any kind of work at a lower payment. Good women were those who followed all the instructions and orders placed by their employer or boss.
7. No wages were given to the child for working 7 to 21 years, keeping them imprisoned. They were granted only 20-22 minutes to have a meal for the whole day.
8. Harsh discipline with monotonous work for women was also waged as abused sexually by the employer. They had to join the mine or work after 1 day of giving birth to a child.

Due to these outrages, social, economic, and political unrest were seen at that moment. The reason behind this was the actual and uncontrolled power of the capitalists.

At the very beginning of the factory production, workers unitedly refused to work below a certain minimum wage and organized a boycott of the manufacturers who didn't pay the minimum rate. They formed and maintained a strong and well-disciplined group. As in 1790, labor organization spread rapidly, upper classes became very uneasy about the power of united workers and the result was the Combination Act of 1799. The Combination Act of 1799 was an act to prevent unlawful combinations of workmen for higher wages, and shorter working hours and also prohibited trade unions and collective bargaining by the workers. Otherwise, all of the trade, manufacturing, commerce, and agriculture of the nation would be destroyed.

Another circumstance was the obstruction of the Speenhamland system in 1795 by the capitalists. It was a relief giving system to poorer ones, employed or not, to a certain minimum living standard. Due to this, labor mobility decreased, which was needed more and more at that time. Rural workers were not interested in working and according to capitalists, it was knocking off urbanization and industrialization could fall. Then capitalists opposed any Govt. aid to the poor.

Conflicts between the landlord and the capitalists

During the 1790s, the wars in which England was fighting had cut off much of its food imports and the price of grains rose substantially. While the money wage rose during that period, the amount of food the worker could buy with his pay declined. But at that time the price of manufactured goods did not rise like that of agricultural products.

After a long series of wars, in 1785, the landlord class utilized all of its social, intellectual, and political power to maintain the Corn Law (Tariffs and other trade restrictions on imported food and corn enforced in the United Kingdom between 1785 and 1846). They diminished the level of import of goods and commodities from abroad during that period and increased the price of domestic products to gain more profits.

After a while, capitalists opposed the Corn Law for two reasons. First, because grains and products made from grains constituted the largest part of the necessary subsistence of labor, the higher price of grains forced the capitalists to pay workers a higher money wage for the workers and their families to survive. This higher wage cut into the capitalist's profit. Thus, the high agricultural price transferred the surplus value created by the workers from the capitalist's profit to the landlords' rents.

Second, by the early 19th century, British manufacturing had become much more efficient than their continental competitors. With lower production time and lower prices of the commodities, the British could make a profit. So, if all tariffs could be abolished and free international trade could be established, British manufacturers could undersell their European competitors.

However, landlords wanted to remain predominantly an agricultural economy to perpetuate their social, political, and Parliamentary power and position to have control over the entire Britain. Whereas, the industrial capitalists were interested in specializing in manufacturing to expand their income and power and to reduce the surplus value that was accruing to the landlords.

As the industrial capitalists had economic dominance but the Parliament was under the control of landlords, in 1815, Corn Law was passed that prohibited all grain importation until certainly high domestic prices had been reached. However, in 1846, Parliament voted for the total abolition of the Corn Laws and raised the final political dominance of the industrial capitalists.

The Theory of Population

Thomas Robert Malthus (1766-1834) was the key figure in analyzing the population statistics. His formulation of the population was a landmark in the history of population theories. He generalized the relationship between population factors and social change. He wrote many books, pamphlets, and essays during his lifetime. His writings can be separated into two periods, each characterized by its overriding social concern and theoretical approach.

During the 1790s and early 1800s, his chief concerns were the unrest of labor and the schemes being advocated by radical intellectuals to restructure society to promote the welfare and happiness of workers. These schemes, Malthus correctly perceived, could attempt to promote the cause of laborers only by eroding the wealth and power of the two classes of proprietors, capitalists, and landlords. Malthus was an outspoken champion of the wealthy, and his theory of population provided the framework within which he defended them.

In 1798 he published 'An Essay on the Principle of Population' as it affects the future improvement of society, with remarks on the speculations of Mr. Godwin, M. Condorcet, and Other writers, generally referred to as the first 'Essay on the Principle of Population'. In 1803, he published a revised edition in which the revisions were so extensive that it was, in reality, a new book. This book is generally referred to as the second 'Essay on the Principle of Population'. He later published 'A Summary View of the Principles of Population'. In the Principles, the most important theoretical basis for his defense of the landlords was his theory of economic "gluts," or depressions.

The abject conditions of the working class and the labor unrest of the late eighteenth century spawned many intellectual champions of the working class. Particularly influential among these were the Frenchman Marie Jean Antoine Nicholas de Caritat, Marquis de Condorcet (1743-1794), and the Englishman William Godwin (1756-1836). It was primarily against the ideas of these two men that Malthus's first Essay was directed.

Condorcet had been an important influence in the first phases of the French Revolution. He advocated two basic reforms to achieve economic equality and security.

First, the precariousness of the incomes of the working poor could be eliminated by the government establishing a fund for the welfare of the aged and of women and children who had lost their husbands and fathers.

Second, the power and wealth of capitalists could be reduced by government regulation of credit. By limiting the amount of credit available to the powerful capitalists and by extending credit to ordinary working people, laborers could slowly become more independent of capitalists, and much greater social and economic equality would result.

William Godwin (1756- 1836) was much more radical than Condorcet. While most British conservatives as well as many classical liberal reformers were bemoaning the natural laziness and depravity of the working class, Godwin argued that the defects of the working class were attributable to corrupt and unjust social institutions. The capitalist society, in his opinion, made fraud and robbery inevitable. Men could not always obtain the necessities because the laws of private property created such great inequalities in society. Justice demanded that capitalist property relations be abolished and that property belongs to that person whom it would benefit most. In Godwin's opinion, it most certainly would not be the government. With economic power went political power.

These two ideas of Godwin's were to be voiced again and again by nineteenth-century socialists:

- (1) Capitalist social and economic institutions, particularly private property relations, were the causes of the evils and suffering within the system.
- (2) The government in a capitalist system would never redress these evils because it was controlled by the capitalist class.

According to Godwin, the solution to this social and economic inequality was to:

1. Ban private property.
2. Abolition of government, the abolition of laws.
3. Provide education for all classes of people.

Malthus's opinion about these reforms:

Malthus's first essay was directed against Condorcet and Godwin. Throughout the first Essay, there were two recurring dominant themes.

The first theme: No matter how successful reformers were in their attempts to alter capitalism, the present class structure of wealth proprietors and poor laborers would inevitably reemerge, class division was the inevitable consequence of natural law.

The second theme: Poverty and suffering were inevitable lot of the majority of people in every society. Furthermore, attempts to alleviate poverty and suffering, no matter how well-intentioned they might be, would make the situation worse, not better.

The Theory of Population:

In his Essay on the Principle of Population (1798) Malthus argued that because of the strong attraction of the two sexes, the population could increase by multiples, doubling every twenty-five years. He contended that the population would eventually grow so large that food production would be insufficient.

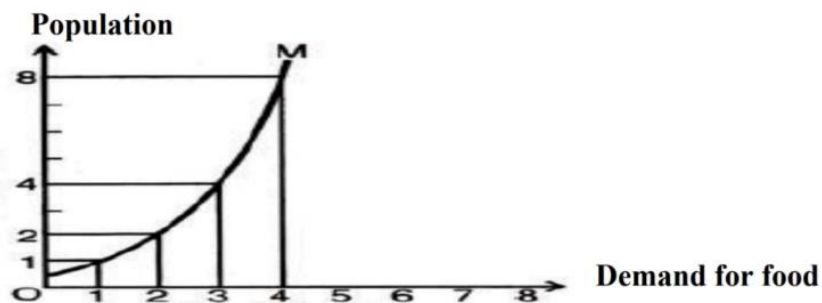
“Population tends to increase at a geometric rate, while the means of subsistence (resources like food, land, etc.) grow at an arithmetic rate.”

Malthus contended that the world's population was growing more rapidly than the available food supply. He argued that the food supply increases in an arithmetic progression (1, 2, 3, 4, and so on), whereas the population expands by a geometric progression (1, 2, 4, 8, and so on). According to him, the population could increase by multiples, doubling every twenty-five years. He said the gap between the food supply and population will continue to grow over time. Even though the food supply will increase, it will be insufficient to meet the needs of the expanding population. Moreover, famine and other natural calamities cause widespread suffering and increase the death rate, which is nature's check against the population.

The food supply in arithmetical progression is measured on the horizontal axis and the population in geometrical progression on the vertical axis. The curve M is the Malthusian population curve

which shows the relation between population growth and increase in food supply. It rises upward swiftly.

Figure 1. Malthusian theory of population



Source: (Malthus, 1798, p. 8)

Malthus based his above arguments on man's two basic characteristics essential to the maintenance of life:

- (i) The need for food, and
- (ii) the passion between sexes.

It was the second which led people to marry at a relatively early age and would result in such a large number of births that the population would double itself in a few years if unchecked by misery and vice. He believed that most people were driven by an insatiable desire for sexual pleasure and that consequently rates of reproduction, when unchecked, would lead to geometric increases in population; specifically, the population would double at each generation. Within a relatively short time, every square foot of the earth would have been inhabited. Thus, the central question to which he addressed himself was what forces had operated to check population growth in the past and what forces were likely to operate in the future. The most immediate and obvious answer was that the population in any given territory was limited by the available food there.

Malthus referred to two types of checks that kept the population down

Positive check: Positive checks increased the death rate; these included famine, misery, plague, war, and the ultimate inevitable check of starvation.

Preventive check: Preventive checks reduced the birth rate; these included such things as sterility, sexual abstinence, and birth control.

The 'positive' and 'preventive' checks that occur in the human population to prevent excessive growth relate to practices affecting mortality and fertility respectively. Malthus saw the tension between population and resources as a major cause of the misery of much of humanity. He was not, however, in favor of contraceptive methods, since their use did not generate the same drive to work hard as would a postponement of marriage. Malthus argued that positive and preventive checks are inversely related to each other. In other words, where positive checks are very effective, preventive checks are relatively less effective, and vice versa. If preventive checks were inadequate, then positive checks were inevitable; and if there was an insufficiency of disease, war, and natural catastrophes, then starvation would always check population growth. Malthus also had a second classificatory scheme, which gets us much closer to an understanding of the normative side of his theory. The positive and preventive "checks that repress the superior power of the population, and keep its effects on a level with the means of subsistence, are all resolvable into moral restraint, vice, and misery.

The population was always checked by some combination of these so that it was kept within the bounds of the available food supply. Within this scheme of classification, Malthus was able to argue that if the wealth and income of everyone in society were increased, the vast majority would respond by having so many children that they would very soon be pushed back to bare subsistence living; only the morally virtuous could escape this fate. "Moral restraint" was defined very simply as "the restraint from marriage which is not followed by irregular gratifications." It is obvious, however, throughout Malthus's writings, that he believed such moral restraint was found only in the persons who had all of the other moral virtues that he valued. It is also obvious that Malthus believed that the lack of sexual restraint would be found among those who squandered every penny that they received above their subsistence on "drinking, gaming, and debauchery.

Thus, in Malthus's theory, the ultimate difference between the rich and the poor was the high moral character of the former and the moral turpitude of the latter. He argued that where moral restraint was absent, the population would be checked by either vice or misery.

Malthus therefore rejected all schemes that would redistribute wealth or income. Such redistributions would merely increase the number of poor workers and push them all back to subsistence. Sometimes Malthus even argued that such a redistribution would not even raise workers' welfare for a brief period until they were able to have children. Malthus also opposed virtually every attempt to legislate some form of relief for the suffering of the poor. The most valuable members of society were, of course, the wealthy class of proprietors, whose value was both economic and cultural. To illustrate the economic value of the wealthy, Malthus argued that in any society the only possible escape from anarchy and total insecurity was the establishment of property rights and marriage. Once these institutions were established, then those persons of high moral character would begin to accumulate, while most of the members of society would dissipate their property in riotous living.

At this point, the lower classes would have no means to continue their existence unless the moral, wealthy elite shared their accumulated funds. But there would be so many poor people that the wealthy elite would have to choose with whom to share their funds. This statement is followed by the previously quoted assertion that the "inevitable laws of nature" decree that all societies will be "divided into a class of proprietors and a class of laborers." The social and cultural value of the wealthy class of proprietors was even greater. Malthus believed that the private property system and the class inequality that it created were responsible for all of the great cultural achievements of humanity. Sometimes Malthus went beyond merely opposing redistributions of wealth and income and legislative attempts to mitigate the harshness of poverty. Perhaps Malthus sensed that even the most hard-hearted conservative might find his policy suggestions a little too harsh, so he ended the first Essay with a sanctimonious appeal to religion and God's will. Malthus's population theory was to have tremendous intellectual influence.

Criticism

The Malthusian theory of population was widely discussed and criticized during the 19th and early 20th. Some of the criticisms are as follows:

1. Mathematical form of the theory wrong: The mathematical formulation of Malthus' doctrine was that food supply increases in arithmetic progression and population increases at geometric progression in 25 years has not been proved empirically. Malthus used the mathematical form to make his principle clear in his first essay but later deleted it in its second edition.

2. Population is not related to food supply but to total wealth: Malthus tried to show the relationship between population and food supply, but the right relationship is between population and total wealth of the country. This is the basis of the optimum theory of population. The argument is that if a country is rich maternally and even if it does not produce enough food, it can feed the people well by importing foodstuffs in exchange for its products or money.

3. Malthus ignored the concept of advanced technology: According to Malthus food supply is not sufficient enough to feed the population. But with the advancement of high technology food production could be increased enough to feed the population

4. Failed to oversee the opening up of new areas: Malthus' theory was narrow and particularly based on the local condition of England. He failed to oversee the opening up of new areas such as Australia, the US, and Argentina where extensive farming led to an increase in food production.

5. Increase in population as a result of declining death rates: Malthus' theory was one-sided. He argued that the population is increasing due to an increase in the birth rate, but due to advanced technology death rate has declined which also caused the population to increase.

Despite these criticisms, it inspired Charles Darwin to formulate his theory of evolution, and variations of this population theory are widely accepted today-especially in theories dealing with economically less-developed countries. The normative orientation of the theory remains now, as with Malthus, to convince us that poverty is inevitable, that little or nothing can be done about it, and that poverty is, generally speaking, due to the weakness or moral inferiority of the poor. (Unat, 2020)

The Labor Theory of Value

As for the class conflict between the proprietors and the capitalist, Malthus argued that for the long-run betterment of the working class and capitalists, the ultimate, long-run interests of both capitalists and laborers would best be promoted by the promotion of the immediate, short-run interest of the landlords. Like Smith, Malthus believed that the quantity of labor commanded was the best measure of value. He also accepted the cost-of-production theory of value. The natural price was the sum of the wages, rent, and profits when each of these costs yielded their recipients the "normal" rate of return on their labor, land, and capital. His discussion of the cost-of-production theory of value differed from Smith's, however, in two very important ways. First, unlike Smith, who saw labor as the only absolutely necessary social cost of production, Malthus argued that wages, rent, and profits were all equally necessary. Second, Malthus did not believe that the market forces of supply and demand would necessarily move the market price toward the natural price. And the difference between the market price and the natural price, according to Malthus, is profit.

To clarify the first point, as to why Smith saw labor as the only absolutely necessary social cost of production, tools and equipment are the product of past labor. Thus, production is seen as a sequence through time of labor exertions directed toward the transformation of natural resources into useful products. This is true for every mode of production. But Malthus shares a notion that in the production process laborers, capitalists and landlords share the same contribution though one can determine the value of product by the quantity of labor embodied in it. Malthus was fond of referring to a "lottery" where some just happened to come into the ownership of only their own labor, while others happened to come into the ownership of capital and land. The fundamental principle of ownership was, he believed, the same in all three cases i.e., because laborers were not owned by others, or were not slaves, they were on the same socioeconomic or legal footing as the owners of the means of production. In other words, owning nothing but one's own labor power was, in principle, no different from owning the means of production. Hence, each class owned a unique type of commodity, and each had to be remunerated if it was to give permission to use its commodity in production.

Malthus's justification of profits as a return to a productive contribution of capitalists was simple. Workers could produce more when they had tools and machinery than when they had none. This added productivity was caused by the capitalists allowing their tools and machinery to be used.

Therefore, Smith was wrong, and capitalists did contribute to production. Malthus entirely missed Smith's point that tools and machinery were simply the present embodiment of past labor exertions. To justify the existence of rent in determining the value of product, Malthus said that rent was not the return to an artificially created restraint on supply; it was due to the differences in nature's gifts to man. He argued that some differences in fertility were due to the landlord's improvement of the soil and thus defended the interests of landlords.¹

¹ More on rent are included in the theory of rent of Malthus.

Theory of Rent of Malthus

Malthus was a spoke person of landlords over his lifetime. This makes clear that he was in favor of rent. In 1815 he published a pamphlet titled '*An Inquiry into the Nature and Causes of Rent, and the Principles by Which It Is Regulated*'. There he developed a similar rent theory as David Ricardo, his lifelong intellectual antagonist, but a close personal friend.

In his pamphlet, he addressed that many Economists of that and previous eras like Adam Smith, the Economists of the school of M. Quesnay considered rent as a common monopoly analyzing its characteristics. Malthus argued that rent can be called a partial monopoly as it is an ordinary excess of the price of raw produce which differs from a common monopoly by nature and laws of government.

In his pamphlet, he provided three causes of the ordinary excess of the price of raw produce above the costs of production-

First, the quality of the soil, by which it can be made to yield a greater quantity of the necessities of life than is required for maintenance of the persons employed on the land.

Second, when essential items are distributed fairly, they can generate their own demand. As the quantity of essential items increases, more people will need and want them.

Third, the comparative scarcity of fertile land, either natural or unnatural.

(Malthus, 1815)

Malthus insisted that "rent is the natural result of a most inestimable quality in the soil, which God has bestowed on Man-the quality of being able to maintain more persons than are necessary to work it." He argued, "Diversities of soil and situation must necessarily exist in all countries. All land cannot be the most fertile."

When a country had a small population, it could grow enough food on the best land available. As the population grew, there was a need to cultivate less fertile land, leading to increased costs. To make farming on less fertile land profitable, prices for agricultural products had to rise. This created a surplus, known as rent, based on the differences in the fertility of the land.

Malthus believed that landowners deserved this surplus because it was a result of 'nature's gifts' to them, not an artificial restriction on supply. He also thought that some differences in fertility were due to the landowner's efforts to improve the soil. According to Malthus, the special value of rent was that it contributed to increased food production, allowing a larger population to be fed.

In contrast, Malthus argued that manufacturing and industry did not create the same benefits because they did not provide for a larger population's necessities.

Malthus concluded that when profits were high, the economy was doing well, and the population was growing, it led to more agricultural production. As agriculture expanded to less fertile land, rents increased. So, according to Malthus, high rents were a sign of overall economic and social success.

While working on rent theory, Malthus also complemented and justified profit as a productive return for capitalists. According to Malthus, workers could produce more when they had tools and machinery from capitalists than when they had none. The addition of productivity was possible due to capitalists who allowed workers to use their machinery. Profits of capitalists were simply the return of this contribution according to him. Malthus did not agree with the 'tools and machinery are contributions of past labors' concept by Adam Smith.

The Theory of Gluts

Along with the theory of value, the theory of gluts is the other theory that Malthus and Ricardo debated about. The theory suggests that an excess of supply over demand is possible, as opposed to the idea of Smith and other classical economists that capitalism would never experience this problem. According to them, the aggregate demand and aggregate supply are to be equal as investment equals savings. But the people at that time saw economic depression where businessmen had always had problems finding buyers for their commodities, productive capacity had gone unused, unemployment among workers had been much worse than usual, and increases in poverty and social distress have been the inevitable results. Malthus wrote the *Principles* with the single goal to promote an understanding of the crises, or gluts and to propose policies to mitigate them.

Malthus searched for the source and remedy for this lack of demand by analyzing the patterns of expenditure of each of the three classes. Laborers spent all of their income to maintain their subsistence level. Capitalists accumulate capital and they do not have the inclination nor the time to spend their profits on consumption or personal services. Landlords, on the other hand, spent all of their income on things that bring comfort to them. They spend on servants and patronize the arts, universities, and other institutions of culture. According to Malthus, all the spending by landlords on consumer goods and personal services brings about an increase in demand.

Malthus gave the sole reason behind gluts as the excessive profits leading to an unsustainable rate of capital accumulation. Capitalists always tended to spend their profits on new capital. They could not always employ more workers and make more profits as they expand their capital. Malthus gave two reasons for that. Either the new capital would embody the same technology as the old or it would embody technical innovations that rendered workers more productive. The first case required sufficient workers to be available and with that, the workforce needed to grow at the same rate as the capital grew. But Malthus pointed out that capital would grow immediately but for the population which in turn the workforce to be available to be employed in this new capital would take at least sixteen or eighteen years. This would raise the problem of unemployment and idle capital. The temporary shortage of labor would raise the wage rate and ultimately the capitalists would prefer to hold the money in cash and stop spending. This will again cause an insufficiency of demand. The second possible type of accumulation involved a technical change that augmented

the productivity of labor. This new labor-saving capital would act as a substitute for labor. The same quantity of produce could be produced with more capital and less labor. However, the displacement of workers would reduce demand.

The way out of this problem is to take policies that would alter the distribution of income. Malthus opined that policies needed to be pursued that would leave capitalists with lower profits and some other class with more income to spend on consumption. By some other class, Malthus once again meant landlords who had both the will and power to consume more material wealth than they produced. They would hire more servants and other unproductive laborers or in Malthus's words providers of "personal services" who would spend their incomes on the material commodities produced in the manufacturing sector. Another way to buy out the idle products in the warehouse or, in other words, generate demand, Malthus suggested that the government should subsidize the landlords or capitalists should give the products to the landlords at a discount rate. In a nutshell, Malthus's solution to the problem of gluts lay in the creation of an army of unproductive workers who were the servants of the landlords. Because they would consume material wealth without producing it thereby eliminating the problem of inadequate aggregate demand. By this solution, Malthus supported corn law for it would permit landlords to receive more rent, and, by that, through their expenditures and those of their servants, would contribute to more aggregate demand without contributing to further increased production.

David Ricardo (1772-1823)

David Ricardo was the son of a wealthy English capitalist who made a fortune in the stock exchange. Ricardo surpassed his father's success and became very wealthy by age thirty. After reading Adam Smith's "The Wealth of Nations" in 1799, he dedicated his time to studying and writing about political economy while continuing to grow his wealth. Ricardo's economic theories, based on abstract deductive models, influenced both radical Marxist and conservative neoclassical traditions for centuries, making him one of the most influential economists. Living in the same era as Malthus, Ricardo was influenced by the French and industrial revolutions, working-class unrest, and conflicts between English capitalists and landlords. Ricardo accepted Malthus's population theory and Malthus's conclusions regarding the nature and causes of laborers' poverty.

After reading Malthus's pamphlet '*An Inquiry into the Nature and Causes of Rent*' in 1815 and acknowledging that Malthus's theory of rent complemented the theory of profits on which David Ricardo was working for some time, Ricardo published 'An Essay on the Influence of a Low Price of Corn on the Profits of Stock, Showing the Inexpediency of Restrictions on Importation' just after three weeks of Malthus's publication. Ricardo developed the initial essentials of his theory of distribution there.

The Theory of Rent & First Approach to Profits

In Ricardo's book named '*On The Principles of Political Economy and Taxation*', he addressed rent as the "portion of the produce of the earth which is paid to the landlord for the use of the original and indestructible powers of the soil." In that book, he recalled Adam Smith's view on rent, he stated that rent is paid to landlords for the liberty of selling the produce, not for the liberty of growing it. (Ricardo, 1815)

Ricardo's theory of the determination of rent was based on two core assumptions-

1. Land differed in its fertility and all lands could be arrayed along a spectrum from the most fertile to the least fertile.
2. Competition always equalized the rate of profit among the capitalist farmers who rented land from the landlords.

Before discussing David Ricardo's theory of rent, we need to understand what **Net Produce** means. Net produce was the total quantity produced minus all of the necessary costs of production, including the replacement of capital used up in production and the wages of workers. So, Net Produce was the total of the Surplus Value. That is,

$$\text{Net Produce} = \text{Surplus Value} = \text{Total Quantity Produced} - \text{Costs of Production}$$

Another fact to clear up is that as Ricardo developed the theory, he believed initially no rent was given to land. In his words from his book, "On the first settling of a country, in which there is an abundance of rich and fertile land, a very small proportion of which is required to be cultivated for the support of the actual population, or indeed can be cultivated with the capital which the population can command, there will be no rent."

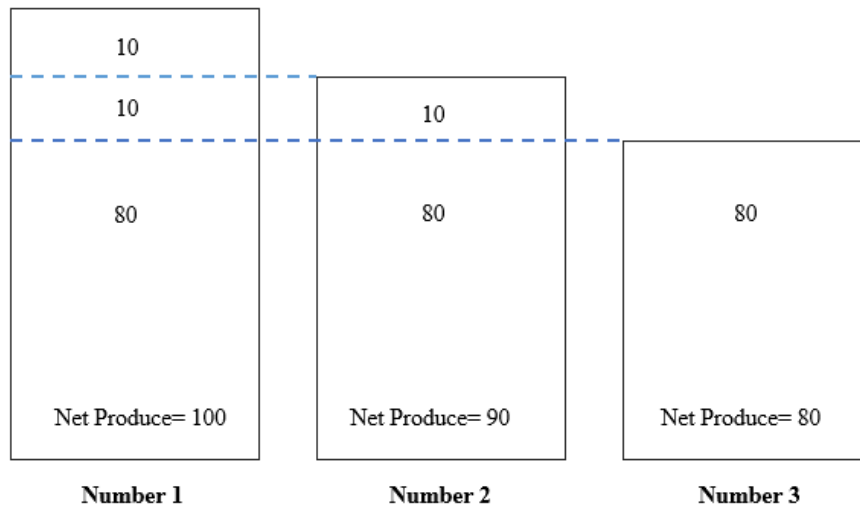
Ricardo's theory of rent is stated here in his own words.

"It is only . . . because land is not unlimited in quantity and uniform in quality, and because, in the progress of population, land of an inferior quality . . . is called into cultivation, that rent is ever paid for the use of it. When, in the progress of society, land of the second degree of fertility is taken into cultivation, rent immediately commences on that of the first quality, and the amount of that rent will depend on the difference in the quality of these two portions of land.

When land of the third quality is taken into cultivation, rent immediately commences on the second, and it is regulated as before by the difference in their productive powers. At the same time, the rent of the first quality will rise, for that must always be above the rent of the second by the difference between the produce which they yield with a given quantity of capital and labor. With every step in the progress of population, which shall oblige a country to have recourse to land of a worse quality, to enable it to raise its supply of food, rent, on all the more fertile land, will rise."

To get a clearer idea let us consider the example given by David Ricardo.

Figure 2: Net Produce and Rent on Three Tracts of Land



In Figure 2, lands are numbered 1, 2, and 3 in descending form based on fertility, that is Number 1 land is the best or most fertile land or first-degree land followed by Number 2 or second-degree land and Number 3 is the worst or least fertile or third-degree land.

Suppose, the net produce of Numbers 1, 2, and 3 land are 100, 90, and 80 respectively. Capitalist farmers of all land use the same amount of capital and labor.

Initially, when only number 1 land was cultivated, there was no rent. So, the profit rate was 100. But as the economy and population progress, second-degree or number 2 land is introduced. Then rent commences to number 1 land. The rent of that land will be simply the difference between these 2 lands' profit which is 10. Then, the rate of profit for both number 1 and 2 land will be 90. Note that, till now number 2 land does not get any rent. If the economy and population progress further, third-degree or number 3 land will need to be introduced. As soon as third-degree land is introduced, rent will commence to second-degree land and rent of first-degree land will increase. Since the net produce of third-degree land is 80, rent for second-degree rent will be $(90 - 80) = 10$ and new rent for first-degree land will be $(100 - 80) = 20$. Thus, the profit rate for each quality land will be equalized at 80 as Ricardo's assumption and third-degree land will be no-rent land till now.

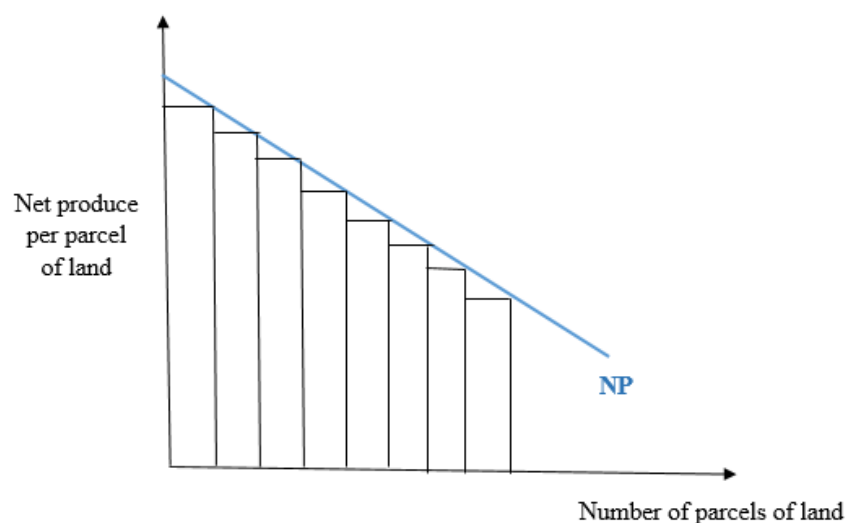
Now, the answer to why first-quality land will get 20 quarters of rent when third-degree land is introduced is competition. Suppose that the farmer on number 1, in Ricardo's example, paid only 15 quarters of rent after number 3 was brought into cultivation. In that event, he would be making

85 quarters of profit (100 quarters net produce minus 15 quarters rent) on the same capital with which the other two capitalist farmers were making only 80 quarters of profit. The other two capitalist farmers could increase their profits by offering to pay landlord number 1 more rent, say 18 quarters, to allow them to farm his land. But as long as the rent on land number 1 was below 20 quarters, capitalists would continue to have an incentive to bid the rent up. Only when the rent was bid up to 20 quarters they would no longer have such an incentive. At that point, the rate of profit would be the same for all capitalist farmers. Ricardo believed that, in general, competition would tend to equalize the rate of profit for all capitalists at the profit rate of no-rent land.

Diminishing Productivity in Agriculture:

Given the assumption of continuously decreasing land fertility, we can divide the land into progressively smaller parcels, with each successive parcel yielding less net produce than the previous one. When represented graphically on a quadrant with axes indicating the number of parcels of land being farmed and the net produce per parcel, the diminishing fertility is illustrated as step-like tops of bars converging toward a simple downward-sloping line. In this representation, each land unit is considered so small that a straight line effectively depicts the diminishing fertility trend.

Figure 3: Diminishing Productivity in Agriculture



In Figure 3, the line labeled NP (Net Produce) serves this purpose, demonstrating that the net produce per small unit of land decreases as the quantity of land under cultivation increases.

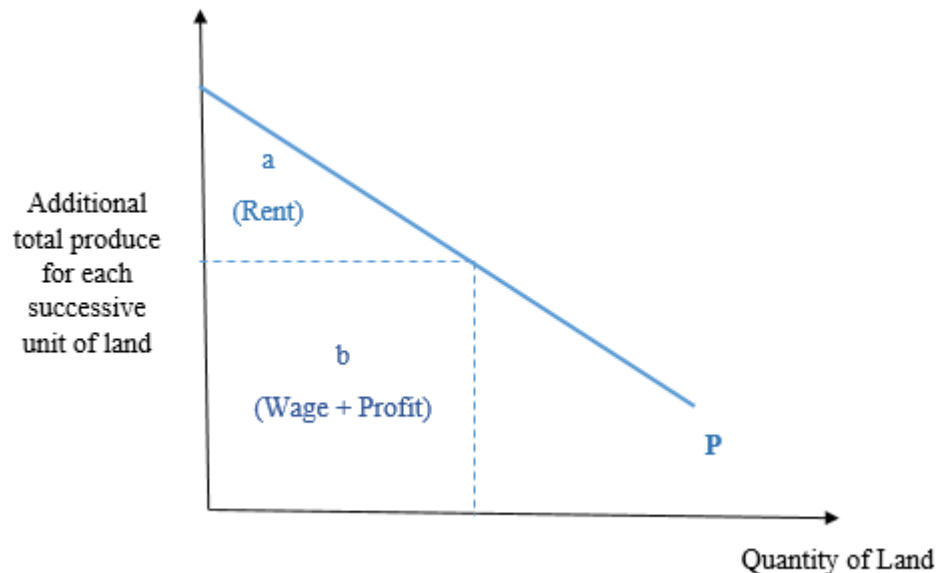
Separation of Rent from Profits and Wages:

If we assume that wages are the only cost of production, then wages paid per unit of land farmed can be added to line NP to show total produce. The result, in Figure 3, is a line, P, showing the total produce for any quantity of land as the amount of land in cultivation is increased. While line NP shows only net produce (profit plus rent), line P shows total produce (profit plus rent plus wages). That is,

$$\text{Net Produce} = \text{Profit} + \text{Rent}$$

$$\text{Total Produce} = \text{Profit} + \text{Rent} + \text{Wage}$$

Figure 4: Separation of Rent from Profits and Wages



In Figure 4, if x units of land are under cultivation, then y will be the total produce on the last (or no-rent) small unit of land brought into cultivation. The area of triangle 'a' will be the total amount of rent received by the landlord class, and the area of rectangle 'b' will be the total profits and wages received by the capitalists and laborers in agriculture.

Ricardo's theory of profits was a crucial part of his overall economic ideas. In simple terms, he looked at a basic economy with landlords, capitalists, and laborers producing only corn. Ricardo saw profits as an extra amount, and in a competitive scenario, all capitalist farmers' profits were considered equal. Therefore, profits would be determined by the determinants of the profit of the capitalist cultivating the no-rent land.

He accepted Malthus's idea that population growth would push wages down to a basic level. So, according to Ricardo, profits on land with no rent were the total corn produced minus the subsistence wages for the workers. In simpler words, profit was what was left after paying wages. In this basic model, the capital was just the corn given as wages.

The rate of profit was the ratio of the extra corn produced on the no-rent land to wages, both measured in corn. This meant that if the extra corn produced per unit of land decreased and wages stayed the same, the rate of profit would go down.

This perspective on profits is often called Ricardo's "**Corn Theory of Profit**". He believed this model could be expanded to include other goods. Essentially, if population growth affected profit rates in agriculture, and if profit rates were determined by productivity on no-rent land, then the same principles applied to the manufacturing sector. In short, according to Ricardo, profit rates in both agriculture and manufacturing depended on productivity on land with no rent.

Economic Basis of Conflict Between Capitalists and Landlords:

On 24th January 1817, in Ricardo's 53rd letter to Malthus, he stated "Surely Buchanan is right and your comment wrong; rent is not a creation but a transfer of wealth. It is the necessary consequence of rent being the effect and not the cause of high price." (Bonar, 1887)

Also, Ricardo's contention in the Essay is that "the interest of the landlord is always opposed to the interest of every other class in the community." These indicate that Ricardo was against rent and that is where the conflicts between capitalists and landlords begin.

Ricardo believed that economic prosperity was linked to the accumulation of capital, a view shared by classical economists. When capitalists earned profits, they could accumulate capital, creating a cycle of economic growth. This accumulation increased the demand for labor, raising the market

wage rate above the natural wage rate (subsistence) and leading to a population increase. As long as profits continued, this cycle could repeat, fostering economic growth, general prosperity, and wages above subsistence. However, the economy ran into difficulty because of diminishing productivity in agriculture, which caused rents to squeeze out profits. That is,

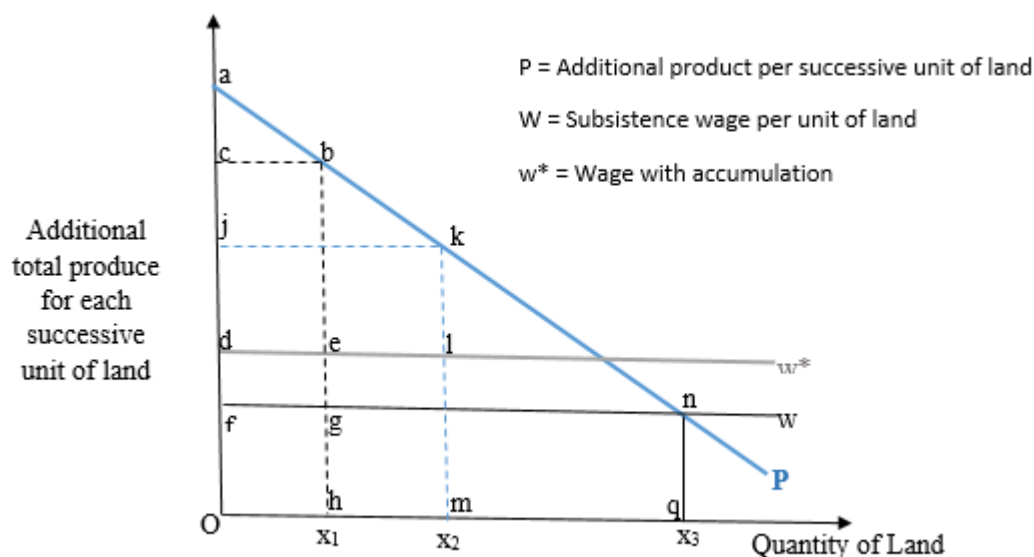
Profit → Capital Accumulation → Demand for Labor ↑ → Market Wage Rate ↑ (Above Subsistence Level) → Population ↑ → The cycle continues until wage falls back to the subsistence level and Profit is squeezed out

Now, in Figure 5, a line labeled w shows the subsistence wage that must be paid to the laborers farming one unit of land and the double line w^* has been added to show the somewhat higher wage that will prevail as long as capital accumulation is occurring.

Before proceeding, we assume-

we observe the economy at a point where x_1 units of land are under cultivation. Also, accumulation has taken place in the past and the wage rate is at w^* .

Figure 5: Changes in the Distribution of Income as the Margin of Cultivation Is Extended



At point x_1 , the entire produce going to rents is represented by the triangle abc . Wages correspond to the rectangle $Ohed$, where $fged$ indicates the excess of wages beyond the subsistence level. Profits are the leftover portion, represented by the rectangle $debc$. That is,

At x_1 ,

Rent = abc

Wage = $Ohed$

Profit = $debc$

With wages at w^* , above subsistence, population growth occurs, necessitating the cultivation of more land. Now, imagine that the population has grown to the point where x_2 is the amount of cultivated land. At this stage, wages are depicted by the rectangle $Oml d$, rent is the triangle akj , and profits are the rectangle $dlkj$. Thus,

At x_2 ,

Rent = akj

Wage = $Oml d$

Profit = $dlkj$

Notably, even though the wage rate has remained constant, both the total profits as a portion of the total product and the profit rate have significantly decreased.

Once the economy has brought x_3 amount of land under cultivation, wages will have been pushed back to subsistence (w); rent will be the area of triangle anf , and wages will be the area of rectangle $Oqnf$. There will be no profits and wages will be back at the subsistence level. So,

At x_3 ,

Rent = anf

Wage = $Oqnf$

Profit = 0

Ricardo believed that diminishing productivity in agriculture would cause profits to be squeezed out steadily by higher and higher rent, a part of revenue already created.

In Ricardo's model, rent didn't directly diminish profits; it was the rising cost of labor due to increased corn prices, the main staple for labor subsistence. Ricardo explained how growing wages shifted a larger portion of the net product from profit to rent. Assuming constant average prices, he argued that when corn and labor prices rose, adjustments would occur to equalize profit rates across the economy. Corn production became less efficient as cultivation expanded, reducing

agricultural profits. However, manufacturing labor productivity remained constant. To equalize profit rates, manufactured goods' prices had to decrease relative to corn. This led to a uniform but lower profit rate. With each expansion of cultivation, the general price level of manufactured goods decreased, causing a decline in profit, hampering economic growth, and reducing overall social welfare.

Due to these reasons, Ricardo was against the corn laws. These laws, by forbidding the import of grains, compelled the agricultural sector to cultivate less fertile land successively. This trend was diminishing profits and, if prolonged, would eventually pause economic advancement. In the discussion on the corn laws, Ricardo's argument was more logically structured compared to Malthus's, even though it didn't impress many Parliament members who were aligned with landlord interests.

However, Malthus found several grounds to attack Ricardo's argument. One of his objections was that profits are influenced by commodity prices, determined by supply and demand. In contrast, Ricardo's profit theory relies on the condition that the overall price of commodities remains constant, and money retains its value, regardless of fluctuations in labor prices. An increase in money wages doesn't provide insight into the profit rate if commodities experience varied price changes – some rising, some falling, and only a few staying the same.

Criticism of the Rent Theory of David Ricardo:

Ricardo argued about the stationary point of an economy due to no profit, subsistence wage, and huge rent. This concept was criticized as

1. Ricardo said, an economy will not grow under this scenario. However, it grew in the 19th and 20th centuries due to technological change.
2. Ricardo only thought about class conflict, not about economic growth or prosperity.
3. Ricardo developed his theory of rent depending on Corn law which is not valid always.

The Labor Theory of Value

Though Ricardo stated in his Principles that all commodities which have value must have utility or otherwise they would not be marketable, utility did not regulate value. As he mentioned, exchangeable value derives from two sources-

1. **Scarcity:** Applied to commodities such as rare statues, pictures, paintings, scarce books & coins, wine of a peculiar quality which are wholly independent of the quantity of labor originally used to produce them & varies with the varying wealth and inclinations of those who are desirous to possess them. These products are quite unimportant in his opinion.
2. **Quantity of Labor Embodied to Them:** Applied to reproducible commodities, which need repeated capital accumulation.

Ricardo did not believe that non-reproducible luxuries had any importance in determining the laws that affect the distribution of the product of the earth among the three classes of the community, hence, they were unimportant in their effect on the accumulation of capital since the accumulation of capital was the principal determinant of the welfare of a country. Consequently, it was only reproducible commodities with which his theory of value was concerned.

Ricardo said, "If the quantity of labor realized in commodities regulates their exchangeable value, every increase of the quantity of labor must augment the value of that commodity on which it is exercised, as every diminution must lower it." Ricardo developed the theory by first stating it as the simplified hypothesis that the prices of commodities were strictly proportional to the labor embodied in them in the production process. He then went to some length to describe how this simple principle would have to be modified due to a variety of special circumstances. He believed that these modifications were fully explainable in a systematic and coherent way and, therefore, did not constitute arguments against the labor theory of value, but rather showed the complexity and realism of the theory. Before making these modifications, Ricardo discussed and then dismissed two objections to the labor theory of value. These were, first, that one could not combine different types of labor having differing skills and differing rates of wages; and second, that the labor theory did not account for the increased productivity made possible by natural resources and capital. To answer the first problem, he said difference in skills among labors will be reflected into

the price of the relevant commodities. As a result, a rise of wages will affect the costs of all commodities in equal proportion, and leave the relative values unaffected (*Ricardo and the 93% Labor Theory of Value*, 2024). As for the second objection, he argued, tools & machinery, were intermediate products of labor (contribution of past labor), created only because they contributed to the ultimate end of producing a commodity for consumption. And natural resources were the objects that labor transformed in production. But they were simply gratuitously there and were not a social cost of production.

However, in analyzing the costs of production that would determine the natural price of a commodity, Ricardo considered only profits and wages. Rents, to his view, were not a component part of the costs that determined prices but, rather, were a residual determined by prices. Thus, the problem for the labor theory of value was to show how the natural prices, each being the sum of wage costs and profit costs, were determined by the labor embodied in the production of commodities.

Price Determination with Differing Compositions of Capital:

Adam Smith, had realized that in order for prices to be proportional to the quantities of labor embodied, it was necessary for the ratio of profits to wages to be the same for every commodity. But because competition tended to equalize the rate of profits on different capitals, then an equal ratio of profits to wages necessarily implied an equal ratio of capital to labor in the production of each commodity. But in real world, the amount of capital per worker differed widely from industry to industry, and it was likely that such differences would always exist. He therefore abandoned the notion that the quantity of labor embodied in a commodity determined its value. Ricardo, then showed that even with differing ratios of capital to labor, the labor theory could be modified to show a systematic connection between the labor embodied in a commodity and the exchange value of that commodity. An example will be convenient to understand this. Let's assume, there are two capitalist firm, the first one employs 100 workers with zero capital to produce good A and the second firm employs same number of workers and extensive machineries to produce good B. Therefore,

The value of the commodity A = wages of 100 workers + 10% profit rate

The value of the commodity B = (wages of 100 workers + 10% profit rate on it) + (wages of workers needed to produce those machineries + 10% profit rate on it)

Let us assume that when these costs are totaled, the price of the second commodity is twice that of the first. Now, in the next production period, say, the wage rate increases. Given the same level of output and the same employment of labor using the same productive techniques, it is obvious that the higher wages will result in a decrease in profits. To derive the new prices, wages constituted roughly 90 percent of the commodity A's cost, and profits constituted 10 percent. Thus, the increase in wages will have a very large effect on the new price, and the decrease in the rate of profit will have a relatively small effect. Commodity A's price will undoubtedly go up substantially. Wages constituted a relatively small percent of the commodity B's cost, so the impact of the increased wages on its total costs will be relatively small. And as we see, profit rate constitutes a large portion of the cost of commodity B, therefore, the effect of the new lower rate of profits will be much greater on the costs of the commodity B than on those of the commodity A. Three possibilities emerge regarding the change in the price of the commodity B -

- a. If the increase in wage cost more than offsets the decline in the profit rate, price will rise, but by a much smaller percentage than the increase in the commodity A.
- b. Increased wage costs may be exactly equal to its decreased profit costs, in which case the price will remain unchanged.
- c. The decrease in profit costs may be greater than the increase in wage costs, in which case the price will decline.

In each of these three cases one fact emerges: whether the price of commodity B goes up or down, the price of commodity A will increase much more than the price of commodity B. Therefore, the price ratio will no longer be 2:1; commodity A will have a relatively higher price. The ratio may go to 1.5:1, for example. However, in this example commodity B must still have a higher value (because it has the same wage costs but also has machinery costs), while the difference between the two prices must decline (because the price of commodity A increases relative to the price of the commodity B). The most important point to note is that the price ratio has changed while the

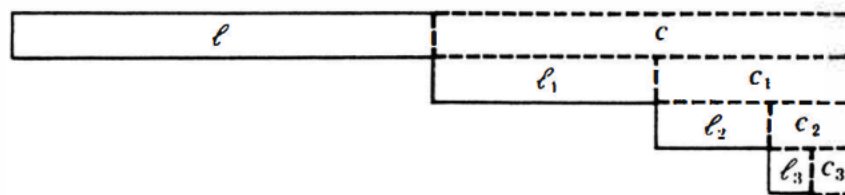
quantities of labor embodied in the two commodities have remained unchanged. There were, three situations in which prices would not be proportional to labor embodied-

1. The ratio of fixed capital to labor differs between two commodities;
2. The durability of fixed capital is not the same between two commodities;
3. The rate of turnover of circulating capital is different.

Each of Ricardo's three cases can be characterized in either of two ways. First, in each of the three cases, the ratio of past labor (embodied in commodities) to present labor is different. Second, as soon as we introduce the words past and present, we introduce a time dimension into production. In Ricardo's three cases, it is also true that if production is seen as a time sequence of labor inputs, then there are differing time sequences of labor inputs in each.

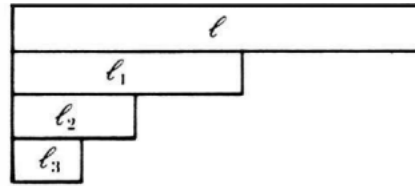
We will now illustrate the effect of differing ratios of capital to labor (or past labor to present labor).

Figure 6: Reduction of capital to past labor



In figure 6, we see how capital can be reduced to a series of time-dated labor inputs. The bars on the top row represent the productive inputs of present labor and previously produced commodities (c). The commodities were produced in the previous period by labor (l_1) and commodities (c_1). These commodities were in turn produced in the period before that by labor (l_2), and commodities (c_2), and so on.

Figure 7: Sequence of time dated labor



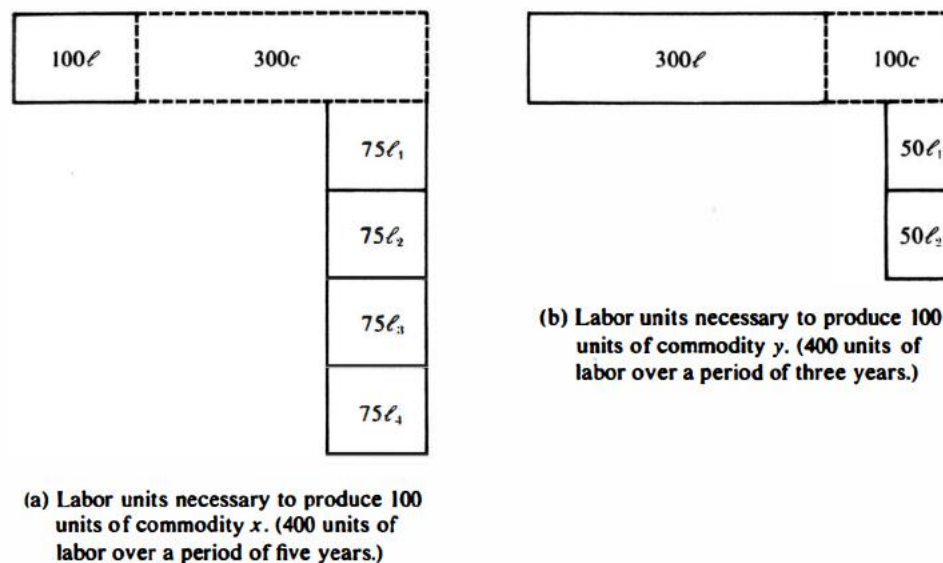
In figure 7, we have simply removed the dotted rectangles representing commodities. This removal reflects the fact that each of the commodity bars has been reduced to prior labor and commodities. At some point, the remaining commodities have become natural resources, which according to Ricardo, have no social cost for its use. All that remains in the production process is a series of time-dated labor. Here, profit is received on capital for the entire period during which the capitalist has funds tied up in the production process. For instance, the capitalist hired labor, represented by 1_3 , three years previously (assuming each period is one year). The labor embodied in the commodities produced by 1_3 does not come to fruition in a finished consumer commodity until the present period ends. Three years ago, the capitalist paid the wages of 1_3 . At the end of that year, the value of the intermediate goods produced by 1_3 was the labor costs plus the profits earned by the capitalist because he or she had money tied up in this production process. At the end of the following year (two years previously), the capitalist again calculated profits, this time on the original wage costs and the profits of the last period that were still tied up in production. The same process is repeated each year, so that the initial costs three years ago grow at a compounded rate each year. Therefore, in Ricardo's terms, either more capital per worker or a longer period of production would give the same result. If the capitalist had \$100 tied up for only two years, the value of that capital at the beginning of the final year would be \$110 (\$100 plus \$10 profit from the preceding year); whereas if the capitalist had \$100 tied up for four years, the value of that capital at the beginning of the final year would be \$133.10 (\$100 plus \$10 profit the first year, plus \$11 profit the second year, plus \$12.10 profit the third year). Thus, the value of capital is greater in the four-year process than in the two-year process, even though the initial wage costs (and hence the original labor embodied in the capital) were the same in each case.

[A Numerical Example of Price Determination:](#)

In figure 3, two production processes have been illustrated. In part (a), 400 units of embodied labor produce 100 units of commodity x, and in part (b), 400 units of embodied labor produce 100 units

of commodity y. It requires 100 units of present labor (labeled 1) and 300 units of past labor (labeled c), applied uniformly at the rate of 75 units per year over the four previous years, to produce x. It requires 300 units of present labor and 100 units of past labor, applied at the rate of 50 units per year over the two previous years, to produce y. If prices were proportional to labor embodied, the two prices would be equal. However, commodity x has both more past labor and a longer period of production; it therefore has a higher ratio of capital to present labor. We assume that all of the commodity inputs are used up at the end of each production process. We assume that the wage rate is \$ 1.00 per labor unit and the profit rate is 50 percent.

Figure 8: Two Production Processes Having the Same Quantity of Labor but Differing Composition of Capital



For (a) in Figure 8,

At the end of the first year, total value produced is $= 75 + 75 \cdot 0.5 = 112.5$

At the end of the second year, total value produced is $= 112.5 + 75 + 187.5 \cdot 0.5 = 281.25$

At the end of the third year, total value produced is $= 281.25 + 75 + 356.25 \cdot 0.5 = 534.375$

At the end of the fourth year, total value produced is $= 534.375 + 75 + 609.375 \cdot 0.5 = 914.0625$

At the end of the fifth year, total value produced is $= 914.0625 + 100 + 1014.0625 \cdot 0.5 = 1521.1$

That is, price of commodity x is \$15.21. Similarly, it can be found that price of commodity y is \$7.31.

In the second, we assume that the wage rate is \$2.00 per labor unit and the profit rate is reduced to 10 percent. Then we find that price of commodity x becomes \$10.62 & price of commodity Y becomes \$9.14. Three important conclusions can be deducted from here –

- a) It is the difference in profits that causes prices not to be proportional to labor embodied. In second case, profits are much smaller, and, hence, the deviation of the price ratio from the labor ratio is also much smaller. If the wage rate would rise to the point where there are no profits, the prices of x and y would be equal, reflecting the equal total labor embodied in them.
- b) If the wage rate increases, the consequent decrease in the profit rate will lower the prices of those commodities in which the profit component of costs is large.
- c) As long as the rate of profit is positive, commodity x will always have a higher price than commodity y.

Ricardo had argued that extending the margin of agricultural cultivation would increase corn prices, that increased corn prices would necessitate higher wages to keep the laborers at subsistence, and that higher wages would always decrease the general rate of profits as well as decrease the average price level of manufactured goods. He had to use the labor theory of value to demonstrate how all of these effects followed from an extension of the margin of cultivation.

Distribution of Income and the Labor Theory of Value

In a simple economy, where corn is the only product produced by labors. The value of corn then is the quantity of labor embodied in the corn production. Here,

Profit = net produced – subsistence wage of labor

But if, population grows and more and more less fertile land is produced, then

Profit = net produced – subsistence wage of labor – rent

In Ricardo's words, of the total value produced by labor, some proportion of what remains of that value, after paying rent is consumed by the producers, and it is this, and this alone, which regulates profits. Ricardo was interested about the total value produced and its division among the three main classes of society.

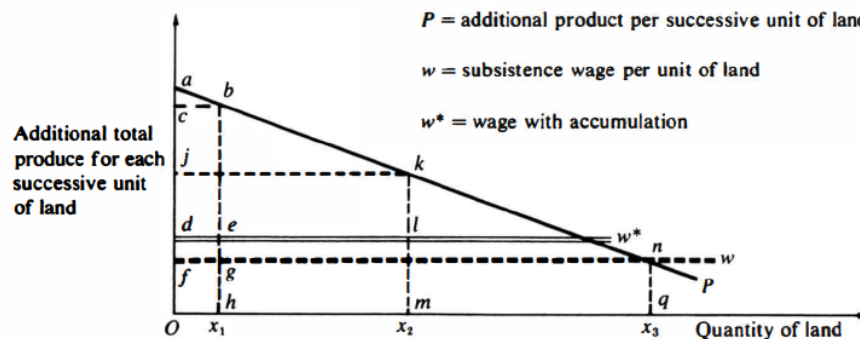
But in economies where corn and manufacturing both products are produced, differences in capital-labor ratio arises and thus variation in prices is seen. These variations, according to Ricardo, were relatively unimportant for two reasons-

- I. Ricardo believed that they would be quite small unless profit rate is more than 50%.
- II. When considering the aggregate quantities, the variations of price ratios from labor ratios would cancel out. Commodities produced with more capital than the social average would have their prices "fall as wages rise, and rise as wages fall," while those with less capital than the social average would have their prices "rise as wages rise, and fall as wages fall." Thus, the variation will be canceled out.

Thus, in the aggregate form, Ricardo showed that labor theory of value can be used for the distribution of income.

But for an individual market in a mixed economy, for instance, to learn the income distribution in the corn market (labor intensive) only or in the manufacturing market (capital intensive) only Ricardo found the value determination difficult, as an invariable measure is needed for this purpose. It is worth mentioning that he didn't find such measure, he thought of gold, but he himself knew that gold cannot be an invariable measure as its value depends on mining process.

Figure 9: Changes in the Distribution of Income as the Margin of Cultivation is Extended



Then he returned to his theory of rent, and said, as more and more land is put under cultivation, profits continue to sink and at x_3 profit will be zero. That means, in the distribution of income, there will be no room for the capitalist. As a result, no capital accumulation will take place which means no reinvestment and economy will stop to grow. Ricardo then blamed the corn law for the justification of his theory. If import of corns are restricted, more corn has to be produced domestically in the less fertile land to support rising population, price of corn will rise in the domestic market and hence wage will rise too to keep workers at subsistence level. As a consequence, portion of rent in the total value produced will rise and profit will be squeezed. The total income will then be distributed between labors and landlords where labors will be at a subsistence level and substantial portion of income will go in the hands of landlords who, according to Ricardo, don't invest in the economy and have no contribution in the production process as lands provided by them is actually given by nature which is not a social cost of production. Again, rise of wage in the corn production will attract labor from the manufacturing sector. To hold labor it will then have to increase wage but as manufacturing sector is capital intensive, a rise in wage will lower its price of produced commodities due to larger effect of reduced profit rate. As a result, both capitalist farmers and manufacturing capitalist will lose due to higher wage caused by the corn law. And the economy will reach a stationary point (no profit, subsistence wage, huge rent).² In particular Ricardo wanted to see the corn laws abolished. He

² (The criticism of stationary point has been already discussed in the criticism of theory of rent).

wanted to show that the interests of the landlords were always opposed to general well-being of society, while those of the capitalists were always in accord with the general well-being of society.

The Impossibility of Gluts

Ricardo opposed what Malthus said to be the solution to the problem of gluts. He argued that subsidizing the unproductive consumption of landlords would be as profitable as having a fire in the capitalist's warehouse that destroyed some of its commodities as it would not contribute to economic growth. To this notion, Ricardo stated "I can see no soundness in the reasons you give for the usefulness of demand on the part of unproductive consumers. How their consuming, without reproducing, can be beneficial to a country, in any possible state of it, I confess I cannot discover" (Dorfman, 1989).

Then Ricardo argued that the forces of supply and demand would automatically adjust prices and the composition of aggregate output. In this regard, Ricardo stated that a general glut would be impossible. This argument is similar to Say's law in its essentials. It suggests that capitalists produce what they do not need only because they intend to exchange it for something that they do need. A producer produces a commodity to exchange for a different commodity. When the producer exchanges the commodity for money he or she intends to exchange the money for other commodities. Here money mediates the exchange but is not desired for itself. Therefore, production creates its own demand. For each dollar in commodities a capitalist produces, he or she has a one-dollar demand for other commodities. This is however a long-run scenario.

Ricardo explained some reasons behind the depression and widespread unemployment that prevailed at that time. He said in the short run crises and depression in the economy can occur due to irrational decisions made by the producers. If a producer mistakenly estimates demand and thus decides to overproduce then oversupply ensues. Additionally, demand for a commodity depends on the taste and caprice of the purchaser, various natural calamities, war, and likewise external forces. Hence, Ricardo stated that the short-run recessions that occurred at that time were due to these abnormal patterns in demand and supply.

Machinery as a Cause of Involuntary Unemployment

David Ricardo wrote a chapter in his *Principles* titled “On Machinery”. In this chapter, he discussed the introduction of new machinery in the production process and its effects. He discussed that the new types of machinery would lower the production cost which would result in increased production and lower prices for the commodities produced by the machinery. The introduction of new machinery comes with a cost which is the displacement of laborers and their job loss. At first, Ricardo believed that the introduction of new machinery would benefit all because he assumed that market price would smoothly and rapidly decline and labor would also be reallocated in the same manner. Later he was convinced that it was not necessarily so. The increase in production would lower the price, this would lower the nominal profit of the capitalists. Because of the drop in price, nominal wage and nominal rent would also decrease. In the whole process, economic welfare is achieved as the economy can produce more by employing less labor. People can buy the same with lower income as real income is not affected. The only backdrop is there would be high unemployment as people would not be employed to the rate people lost employment due to the injection of new machinery in the economy. This would last until a new job is created as a result of new machinery. Ricardo agreed with Malthus that the market might not be very effective in reallocating resources when a change in the conditions of production took place and that the result could be a chronic depression in the labor market that would reduce the total production of the economy. It also meant that Ricardo's faith in the accumulation of capital as the principal force that would increase the economic welfare of all society was unfounded.

The Theory of Comparative Advantage and International Trade

Comparative advantage is a key principle in international trade and forms the basis of why free trade is beneficial to countries. In economics, a comparative advantage occurs when a country can produce a good or service at a lower opportunity cost than another country. The theory of comparative advantage shows that even if a country enjoys an absolute advantage in the production of goods, trade can still be beneficial to both trading partners. This leads to a more efficient allocation of resources globally, increased productivity, and overall higher levels of consumption and economic welfare for both trading partners.

The theory of comparative advantage is attributed to political economist David Ricardo, who wrote the book *Principles of Political Economy and Taxation* (1817). He used the theory of comparative advantage to argue against Great Britain's protectionist Corn Laws, which restricted the import of wheat from 1815 to 1846. In arguing for free trade, the political economist stated that countries were better off specializing in what they enjoy a comparative advantage in and importing the goods in which they lack a comparative advantage.

He was the first economist to argue consistently that free international trade could benefit two countries, even though one country produced all of the traded commodities more efficiently than the other. He was also one of the first economists to argue that, because capital was relatively immobile between nations, a separate theory of international trade was needed.

Ricardo argued that a country need not have an absolute advantage in the production of any commodity for international trade between it and another country to be mutually beneficial. Absolute advantage meant greater efficiency in production or the use of less labor in production. Two countries could both benefit from trade if each had a relative advantage in production. The relative advantage was meant simply that the ratio of the labor embodied in two commodities differed between the two countries, such that each country would have at least one commodity where the relative amount of labor embodied would be less than that of the other country.

It is calculated by finding the opportunity cost for a set of goods. Suppose two neighboring countries produce two sets of similar goods. So to find out the comparative advantage for those two goods, we need to find out the opportunity cost for producing one good over the other good

as the number of skilled labor is the same. Comparative advantage is calculated as Comparative Advantage Formula = Quantity of Good A for Country X / Quantity of Good B for Country X

This formula will help us calculate the opportunity cost for product A; similarly, we need to calculate the opportunity cost for product B. We will do that for both countries. We will be able to determine the comparative advantage of a particular good for a country compared to others by looking at the product of the formula.

David Ricardo famously showed how England and Portugal both benefit by specializing and trading according to their comparative advantages.

Table (a) is a reproduction of Ricardo's example to illustrate the principle of comparative advantage.

Table (a): Number of Hours Required to Produce One Unit of Cloth and Wine in England and Portugal

	Cloth	Wine	Ratio of the price of wine to the price of cloth	Ratio of the price of cloth to the price of wine
England	100	120	1.20	0.83
Portugal	90	80	0.88	1.12

In this table, Portugal has an absolute advantage in the production of both wine and cloth; that is, fewer labor hours are required to produce either commodity in Portugal than are required in England. If we assume that the prices of wine and cloth are proportional to the labor embodied in them in both England and Portugal, then the ratios of the two prices in each country will be the same as the ratio of labor hours required to produce the commodities in each country.

In Portugal, it takes ninety hours to produce one unit of cloth and eighty hours to produce one unit of wine. This means that wine requires only 88 percent as much labor as cloth and that the price of wine is only 88 percent that of cloth. In England, the labor embodied in and the price of wine

are both 120 percent that of cloth. Thus, Portugal uses relatively less labor to produce wine, and the price is relatively lower. On the other hand, Portugal uses 112 percent as much labor to produce cloth as wine, and the price of cloth in that country is 112 percent that of wine. England uses only 83 percent as much labor to produce cloth as wine, and, again, the price of cloth is only 83 percent that of wine. Thus, England uses relatively less labor to produce cloth, even though it uses more labor; therefore, England has a relative advantage in producing cloth.

Now suppose that the Portuguese have been producing only wine (the commodity in which they have a comparative advantage) and decide that they want cloth as well. They have two ways of getting the cloth—diverting some labor from the production of wine to the production of cloth or trading wine with England for cloth. Suppose that if trade takes place it does so at the price ratio prevailing in England. To produce one unit of cloth will require ninety working hours. This means that the Portuguese must discontinue producing 1.12 units of wine for every unit of cloth they produce. But if they trade with England at the English price ratio, they need only give up 0.83 units of wine for each unit of cloth. Trading would leave them a greater combined total of wine and cloth than would the production of both commodities.

Similarly, if England had been producing only cloth but could trade at the price ratio prevailing in Portugal, then it should not have produced wine. To do so would require that it give up 1.2 units of cloth for each unit of wine produced, while it could give up only 0.88 units of cloth for one unit of wine if it trades with Portugal.

Both countries could benefit if each could trade at the price ratio prevailing in the other country. But both countries could benefit if they trade at a price ratio that is between the ratios prevailing in each country. The ratio of the price of wine to the price of cloth is 1.2 in England and 0.88 in Portugal. If both countries trade on a one-to-one basis, one unit of cloth for one unit of wine, both could consume a larger combined total of wine and cloth than if they each produce both commodities.

Indeed, Ricardo's theory of comparative advantage forms the basis of his argument in favor of free trade. He believed that by allowing countries to specialize in the production of goods and services

where they have a comparative advantage, and by engaging in trade without restrictions, the overall wealth and well-being of nations would increase. Ricardo argued that restrictions on trade, such as the corn laws, would limit the potential benefits of specialization and exchange, ultimately reducing the total "sum of enjoyments" available to society. Therefore, his advocacy for free trade was not only an economic argument but also a moral and philosophical stance aimed at promoting greater prosperity and human welfare.

Social Harmony and Class Conflict

The utility theory of value, or any approach to economics that tends to equate prices and utility, usually supports the view that social harmony prevails through the "invisible hand" of the free market. When Ricardo asserted that free trade would increase the "sum of enjoyments" of each country, he was merely restating Adam Smith's principle that free exchange increases the utility or "enjoyments" of both parties to the exchange. When this principle becomes the focal point of economic analysis, the remedy to nearly all problems of human material deprivation becomes one of extending the market or making exchange and trade freer. If this policy is followed, it appears as if everyone would benefit, and hence all interests would be harmonious.

For Ricardo's conclusion to follow from his premises, he had to assume that if England imported the relatively more expensive commodity, then the higher price of that commodity would be a reasonable index of the increase in the "sum of enjoyments." In other words, suppose wine was drunk only by landlords and capitalists and that laborers did not have enough clothing to keep themselves warm. The labor theory perspective tends to focus on the circumstances that resulted in so much of the production of labor being expropriated as rent and profit. The utility theory, however, generally assumes the laws of property ownership and the distribution of wealth to be fixed, or "natural," and, consequently, would tend to focus on the fact that capitalists and landlords would prefer having their surplus in both cloth and wine rather than in cloth alone. Free trade would increase the enjoyment of those who had wealth and purchasing power but not necessarily the enjoyment of the working class.

Ricardo could have suggested, for example, that the importation of wine be prohibited until every worker had adequate clothing. Some would certainly argue that this would increase social welfare more than importing more wine for the capitalists and the landlords would. But he did not do so, instead insisting that free trade would maximize the "sum of enjoyments".

Appears that Ricardo, much like Adam Smith, focused on the labor theory of value and examined class conflicts in his work. He analyzed how factors like wages and profits interacted and often found conflicting interests between different classes, such as landlords, consumers, and manufacturers.

Smith saw capitalism as the highest stage of social development, with property and wealth distribution considered fixed. Ricardo, however, viewed these aspects as eternal and natural, leading to a focus on conflict in his labor theory of value. This conflict implies change, suggesting that socioeconomic systems have a life cycle, akin to growth, maturity, and decline.

Within the context of this ahistorical view, it is not surprising that Ricardo did not consider policies that would increase the "sum of enjoyments" by altering the distributions of wealth, privilege, or power. When these distributions are accepted as natural or eternal, the enjoyment of those who have the purchasing power generally is thought to be synonymous with the overall social welfare. From this, a belief that the free market acts as an invisible hand to maximize everyone's welfare nearly always follows. Thus, Ricardo wrote the following defense of free trade:

"Under a system of perfectly free commerce, each country naturally devotes its capital and labor to such employment as are most beneficial to each. This pursuit of individual advantage is admirably connected with the universal good of the whole. By stimulating industry, by rewarding ingenuity, and by using most efficaciously the peculiar powers bestowed by nature, it distributes labor most effectively and most economically while, by increasing the general mass of productions, it diffuses general benefit, and binds together, by one common tie of interest and intercourse, the universal society of nations throughout the civilized world "

The invisible hand operated locally, nationally, and internationally, harmonizing everyone's interests. That means the notion of the invisible hand suggests that self-interest can lead to collective harmony, operating at various levels from local to international. However, Ricardo's

emphasis on capitalist interests, rooted in the labor theory perspective, led to contradictions regarding whether capitalism fosters conflict or harmony. Instances, such as unequal gains in dealings and trade imbalances between mother countries and colonies, challenge the universality of the invisible hand's operation, highlighting potential conflicts of interest.

Every argument that Ricardo gave showing how coercive restrictions in the free market could benefit the mother country at the expense of the colony could just as easily demonstrate how the exchange between capitalists and laborers could be regulated by coercive restrictions that benefited the capitalist at the expense of the laborer (or vice versa, as many of the leaders in the rising labor movement of Ricardo's time hoped).

Adam Smith and David Ricardo indeed recognized the potential for the government to serve the interests of the wealthy. However, they argued that in a properly functioning market system, competition and self-interest would lead to optimal outcomes for society as a whole, benefiting both the rich and poor. They believed that the invisible hand of the market would naturally regulate economic activity, promoting efficiency and prosperity. Additionally, they advocated for certain governmental interventions, such as enforcing contracts and preventing monopolies, to ensure fair competition and mitigate the exploitation of the poor by the rich. (Hunt, 2011)

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